

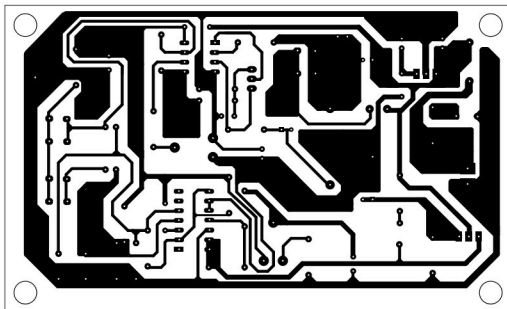


Fig. 13: An actual-size, single-side PCB for the over-load cum low-battery load-disconnect circuit shown in Fig. 6

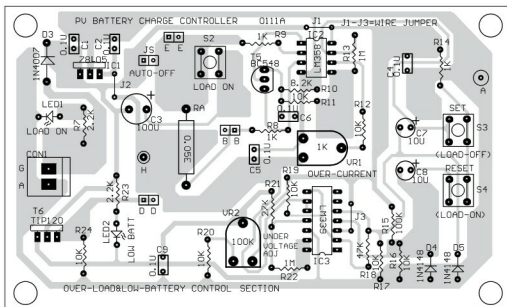


Fig. 14: Component layout for the PCB in Fig. 13

Hence the wired-OR output of comparators N2 and N3 is passed through Darlington transistor T5 for driving MOSFET T4 connected at point D.

PWM charging circuit. Commonly available PWM controller IC TL494, which is widely used in PC SMPS and costs less than an ordinary op-amp, is used for this circuit.

PWM on/off circuit. Error amplifier-2 is employed as enable/disable control to cut off the PWM output

of TL494 when the battery voltage reaches the upper set limit of 14.3V (13.8V+0.5V) and enable it again when the battery reaches the lower set limit of 13.3V (13.8V-0.5V). This function is realised with the help of the circuit shown in Fig. 8. The first comparator provides the necessary cut-off/cut-on points, while the second comparator is used as a simple inverter. Necessary formulae and parameters used for design are shown in Fig. 8 itself.

This circuit is interfaced to error amplifier-2 of TL494 after integration with temperature compensation.

Temperature-dependent reference voltage generation circuit. A careful look at Fig. 8 shows that V_{REF} has been obtained from a temperature-dependent reference voltage (9.2V at 25°C) and one-third of the average V_{BATT} value has been used for comparison with V_{REF} . Variation of V_{REF} shifts the upper and lower threshold points used for switching on/off the PWM output. The circuit used for generation of temperature-dependent reference voltage is shown in Fig. 9.

We had earlier stated that for a 12V lead-acid battery, a 30mV/°C negative temperature correction is required to be applied for the battery voltage set points as shown in Fig. 2. The average charging voltage at 25°C is indicated as 13.8 volts. This means that the average charging voltage at 0°C should be higher by 0.75V (30mV × 25) or it would be 14.55V (13.8+0.75) at 0°C.

National Radio's LM-35DZ temperature sensor IC provides 10 mV/°C. It is one-third of 30 mV/°C

required for correction of the 12V battery. Thus we use one-third of the battery voltage (14.55/3=4.85V) as reference voltage at 0°C and subtract from it 10 mV/°C using a DC difference amplifier formed from LM358 IC as shown in Fig. 9. The 4.85V is derived from 5V reference (generated by TL494 IC) using a resistive divider. This circuit provides a temperature-corrected reference voltage of 9.7V at 0°C (or 9.2V at 25°C and so on). This